

HORTENSE LE

Machine Learning Scientist — Computational Pathology & Medical Imaging

New York, NY · hortensemai25@gmail.com

[LinkedIn](#) · [Google Scholar](#) · [GitHub](#) · [ORCID](#)

Machine learning scientist with 5 years at NYU Langone Health building diagnostic and prognostic models from histopathology, CT, and MRI. Works with pathology and CT foundation models on multicenter cohorts with external validation, and owns projects end to end, from cohort assembly to publication. 20+ publications, 260 citations, h-index 7, including first-author work in *Nature Reviews Clinical Oncology* and *Scientific Reports*, plus open-source releases of the survival and phenotyping pipelines.

EXPERIENCE

Data Scientist — NYU Langone Institute for Computational Medicine

Jan 2021 – Present

New York, NY

- **Colorectal cancer survival — technical lead.** Built an interpretable disease-free survival model on H&E whole-slide images across a multicenter cohort of **1,024 patients from 16 institutions**, validated on held-out external centers (**C-index 0.68**). Benchmarked self-supervised phenotype clustering against pathology foundation models (UNI, CONCH, TITAN, H-optimus) with attention-based multiple-instance learning; every risk score traces back to the tissue morphologies driving it. Preprint under review; pipeline released as [PanColon-CHIPS](#) and [SurvCLAM](#).
- **Kidney cancer imaging — project lead, ongoing.** Predicting tumor aggressiveness and post-nephrectomy renal function from preoperative CT in **469 patients**, benchmarking radiomics, 2D/3D CNNs and ViTs, tabular models, and CT foundation models (Merlin, RenalCLIP) on nnU-Net/TotalSegmentator segmentations. Separately automated contrast-phase labeling at **macro-F1 0.93** across **542 scans**, removing a manual review step that had gated every study in the program.
- **Diagnostic models across three disease areas.** Self-supervised models on bone marrow biopsies separating myelodysplastic neoplasms from clinical mimics — a distinction that normally requires cytogenetics (**AUC 0.78**; co-first author, *Blood Neoplasia* 2026). Histomorphological phenotype clusters from unannotated H&E linked to immunotherapy response in lung adenocarcinoma (first author, *Scientific Reports* 2025; code at [cn_HPL_LUAD](#)). Quantitative late-gadolinium-enhancement features on cardiac MRI for sarcoidosis diagnosis (first author, *J Cardiac Failure – Intersections* 2025).
- **Pipelines and infrastructure.** Containerized training and inference in Docker on SLURM GPU clusters, handling **10+ TB** of whole-slide and volumetric CT data. Assembled cohorts from the urology data warehouse in Databricks/SQL. Replaced brittle regex parsing with LLM pipelines (MedGemma, GPT, Claude) that extract structured variables from radiology reports and operative notes across heterogeneous dictation styles, validated against clinician-reviewed records. Shipped an internal web app that clinical collaborators use to inspect model outputs on their own cases.
- **Collaboration. Sole data scientist on every study above**, working directly with pathologists, radiologists, and surgeons from study design through manuscript. Mentored a postdoc, a PhD student, and a master's student; contributed methods and preliminary results to grant proposals.

Research Fellow — Heffner Biomedical Imaging Lab, Columbia University

May – Dec 2021

New York, NY · concurrent with the role above

- CNN for frequency-and-phase correction in MEGA-PRESS MR spectroscopy, published in *Magnetic Resonance in Medicine* (2022); also modeled pulmonary blood volume maps from CT.

Graduate Research Assistant — Institut Pasteur, Paris · UC San Diego

2019 – 2020

- Benchmarked Faster R-CNN, Mask R-CNN, and U-Net for immune cell detection in NSCLC with Oslo University; EEG analysis across 80+ subjects, contributing to a *NeuroImage* publication.

SELECTED PUBLICATIONS

20+ total — full list on [Google Scholar](#)

- **Le H**, Tsirigos A. AI accurately identifies targetable alterations in lung cancer histological images. *Nature Reviews Clinical Oncology*, 2025. (first author)
- **Le H**, Coudray N, Yeaton A, et al. Contrastive learning uncovers cellular interactions and morphologies in the tumor microenvironment of lung adenocarcinoma linked to immunotherapy response. *Scientific Reports*, 2025. (first author)
- Mehrtash V, **Le H**, et al. Self-supervised learning can distinguish myelodysplastic neoplasms from clinical mimics using bone marrow biopsies. *Blood Neoplasia*, 2026. (co-first author)
- Claudio Quiros A, Coudray N, Yeaton A, Yang X, Liu B, **Le H**, et al. Mapping the landscape of histomorphological cancer phenotypes using self-supervised learning on unannotated pathology slides. *Nature Communications*, 2024.

SKILLS

Modeling — Self-supervised learning, multiple-instance learning, pathology and CT foundation models (UNI, CONCH, TITAN, H-optimus, Merlin, RenalCLIP), 2D/3D CNNs and Vision Transformers, VLMs, radiomics, survival analysis

Imaging — Whole-slide histopathology, CT and MRI segmentation (nnU-Net, TotalSegmentator), multiplexed imaging, spatial transcriptomics

LLMs — Clinical text extraction (MedGemma, GPT, Claude), structured output validation

Engineering — Python, PyTorch, SQL, R; Docker, Databricks, SLURM/HPC, GPU clusters, Linux

EDUCATION

MS, Biomedical Engineering — Columbia University · *Bioinformatics and machine learning*

2022

MS, Biomedical Engineering — Université de Paris / PSL / Arts et Métiers · *Biolmaging*

2020

BS, Physics-Biophysics — University of California San Diego

2018